

26TH ANNIVERSARY OF
PCI-IOSS
SPINE SURGERY SUMMIT
in conjunction with
ISASS AP 2024

INTERNATIONAL SOCIETY FOR THE ADVANCEMENT OF SPINE SURGERY ASIA-PACIFIC SECTION

28 - 31 AUGUST 2024 | YOGYAKARTA, INDONESIA

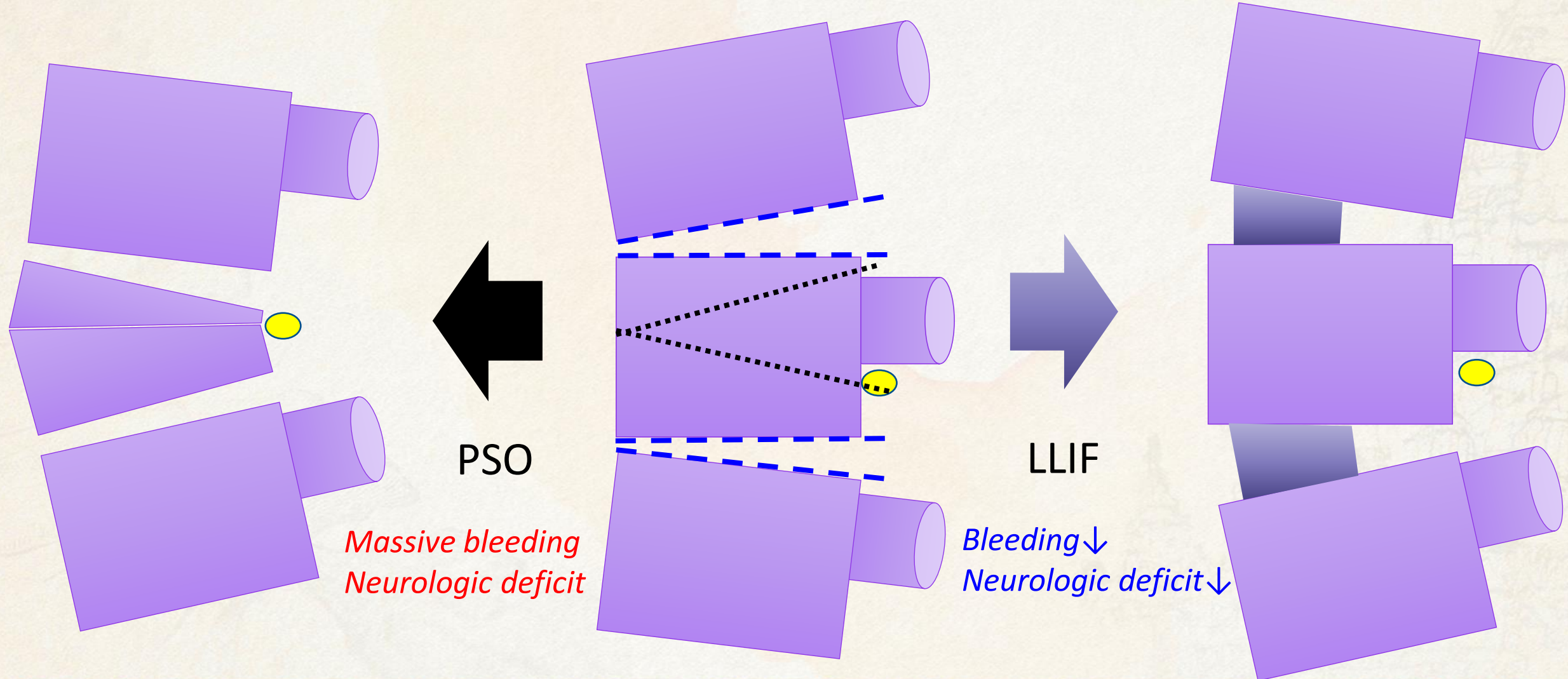
ACR as a minimally invasive surgical
technique to treat severe flatback
deformity

Se-Jun Park
Samsung Medical Center
Seoul, Korea

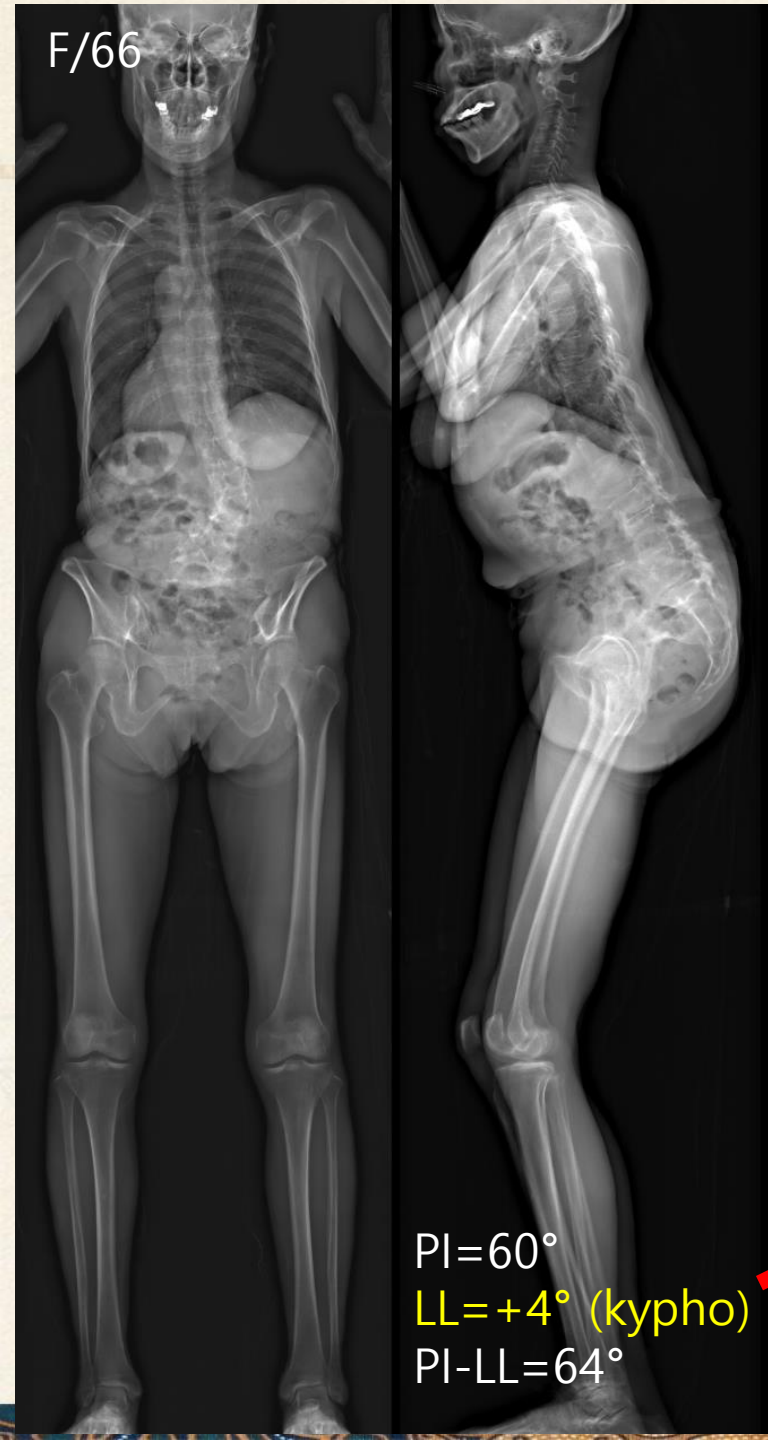


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How to correct severe sagittal malalignment?



F/66



PI=60°
LL=+4° (kypho)
PI-LL=64°

DONESIA



32.4°

Conventional LLIF surgery is enough for flexible deformity

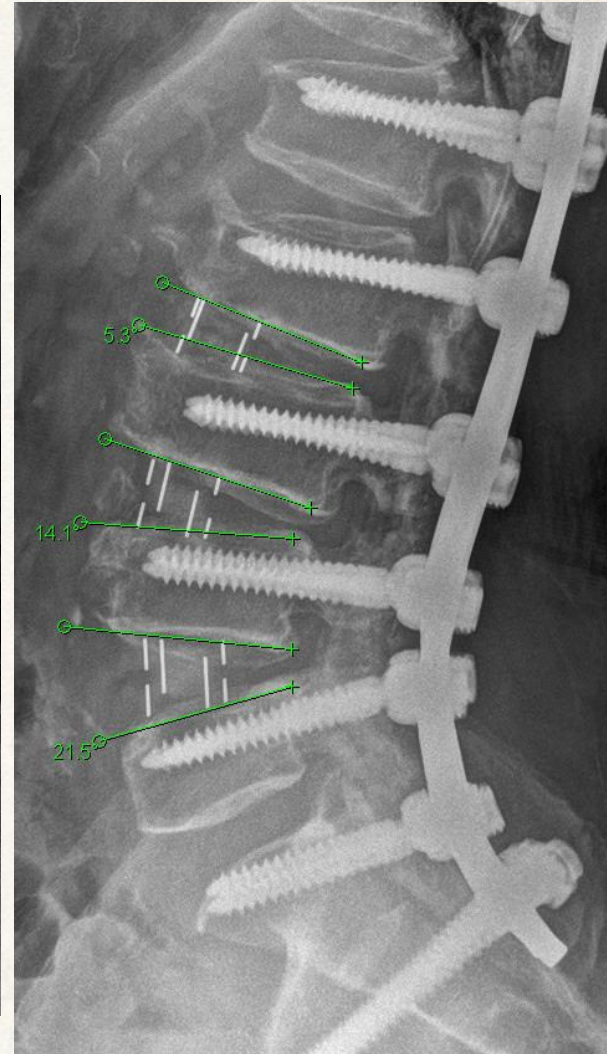
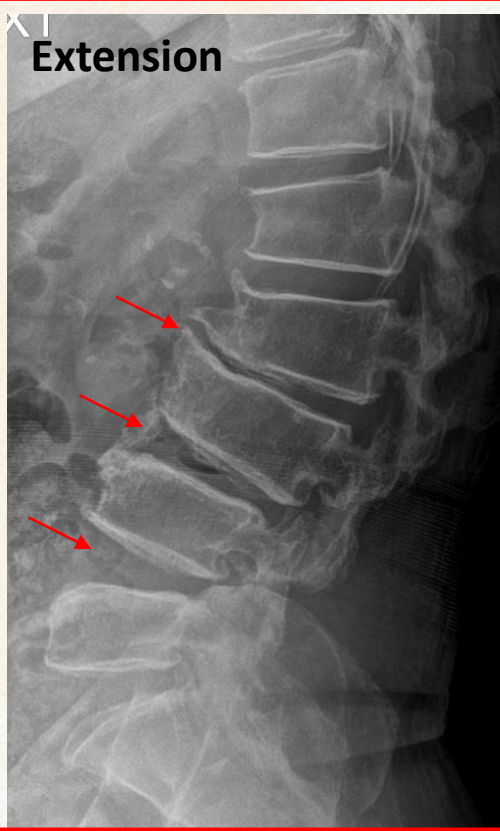
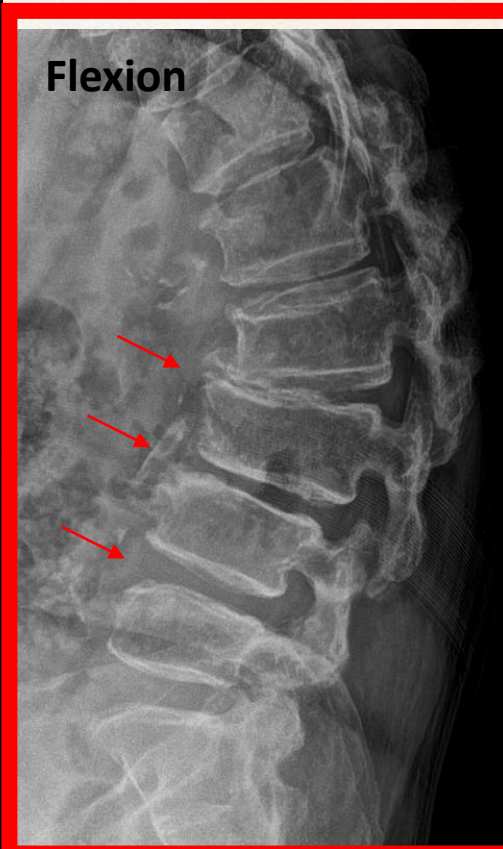


PI=60°
LL=53°
PI-LL=7°

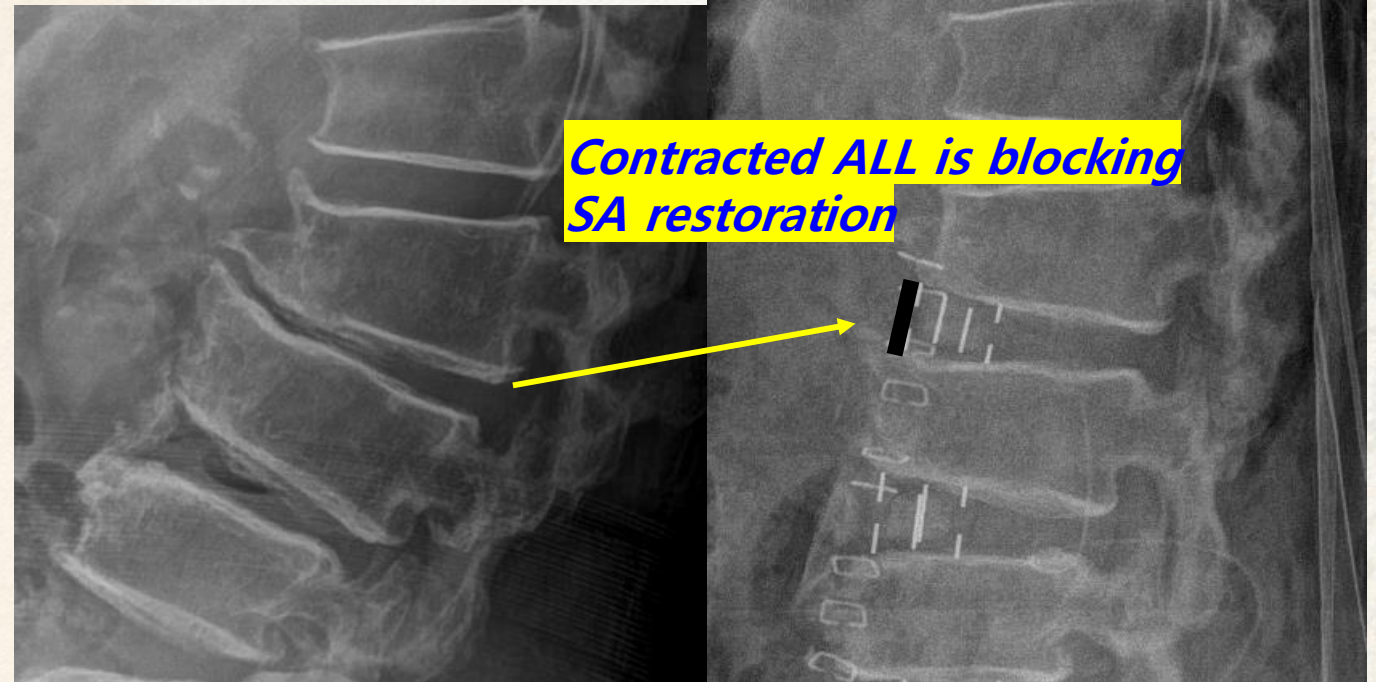
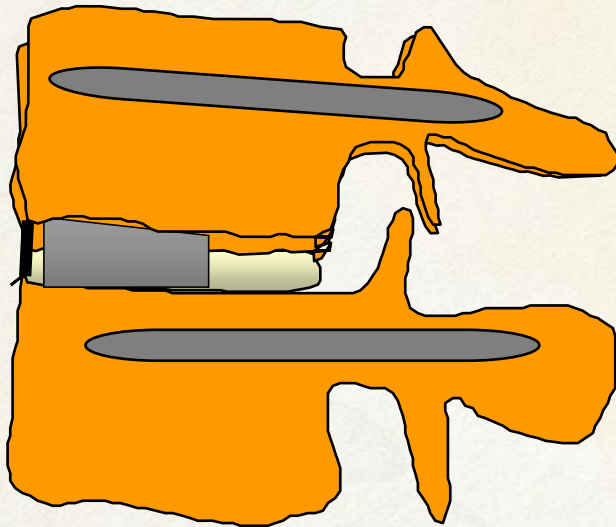
F/74
Severe back pain
with stooping

Seg. lordosis by LLIF $\propto$ Preop Seg. flexibility

Flexibility : L4-5 > L3-4 > L2-3

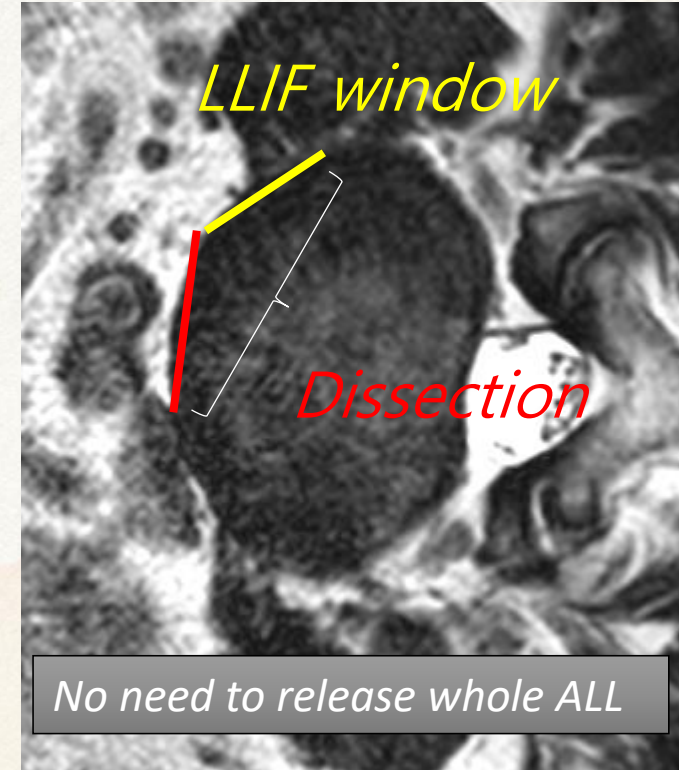
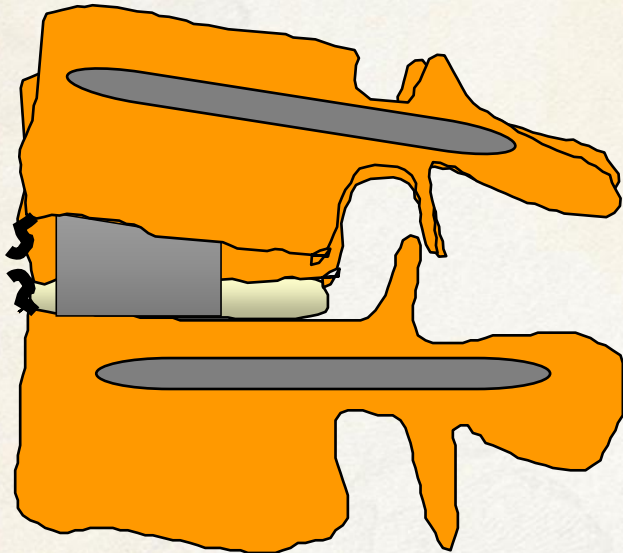


LLIF in non-flexible segment



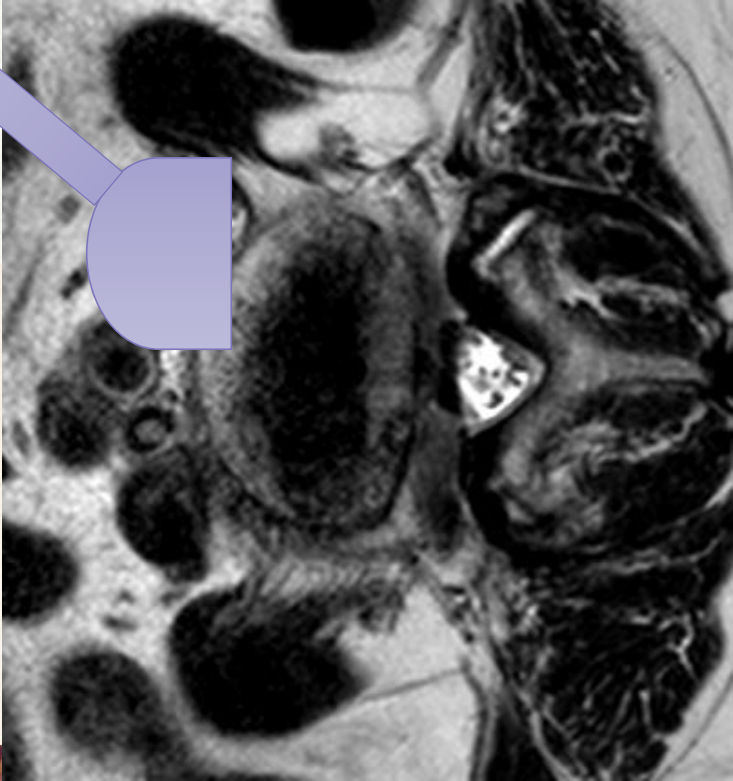
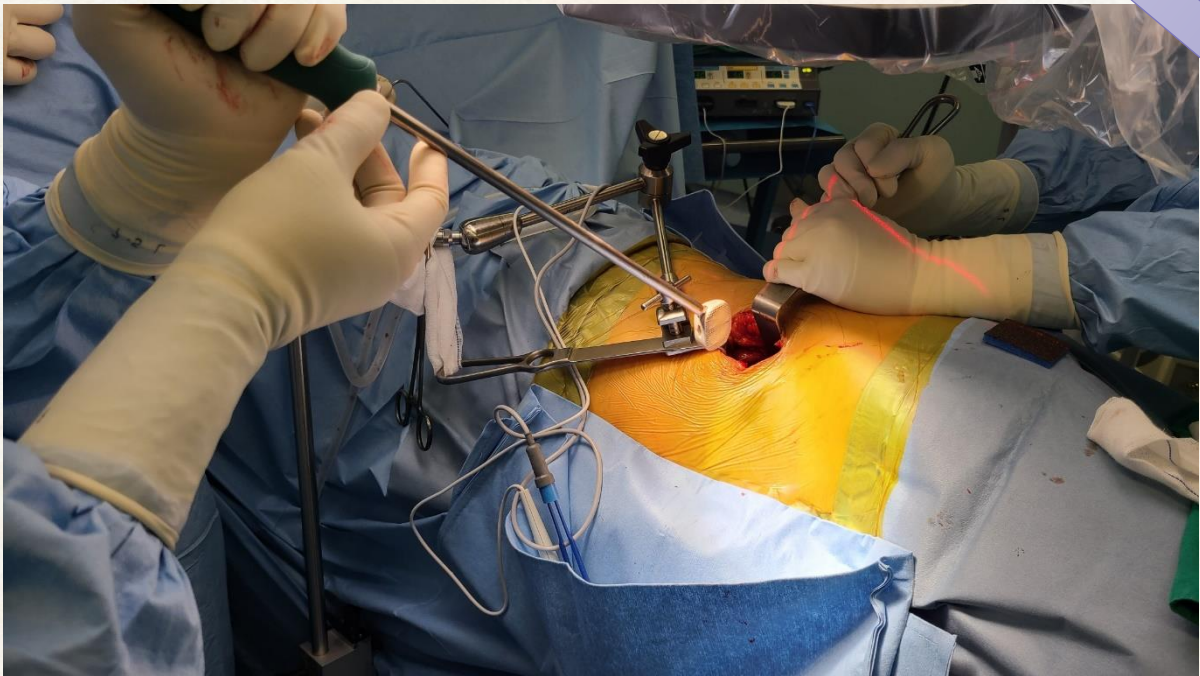
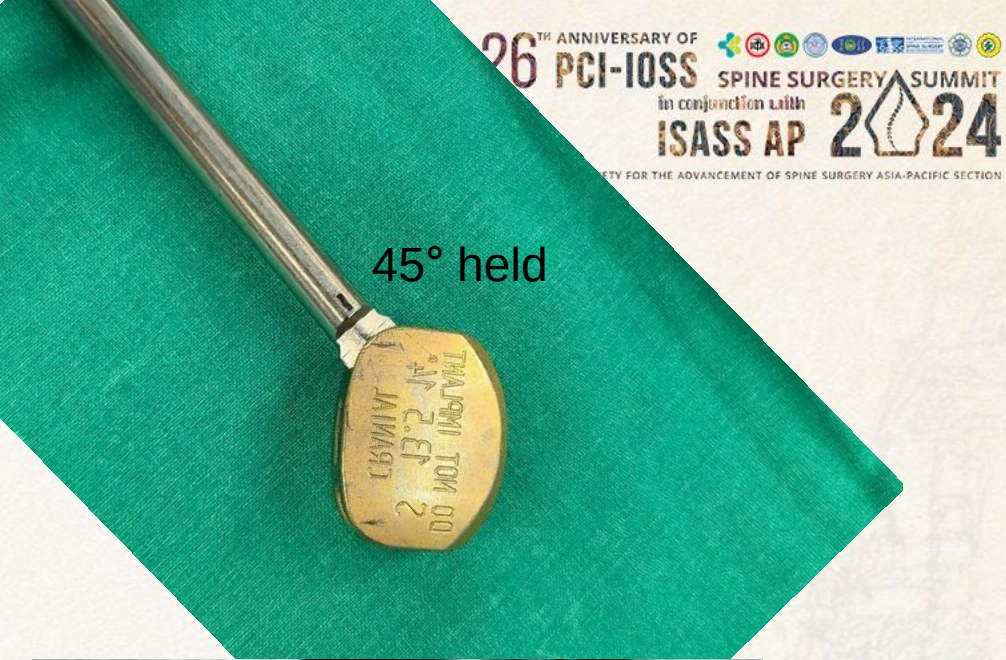
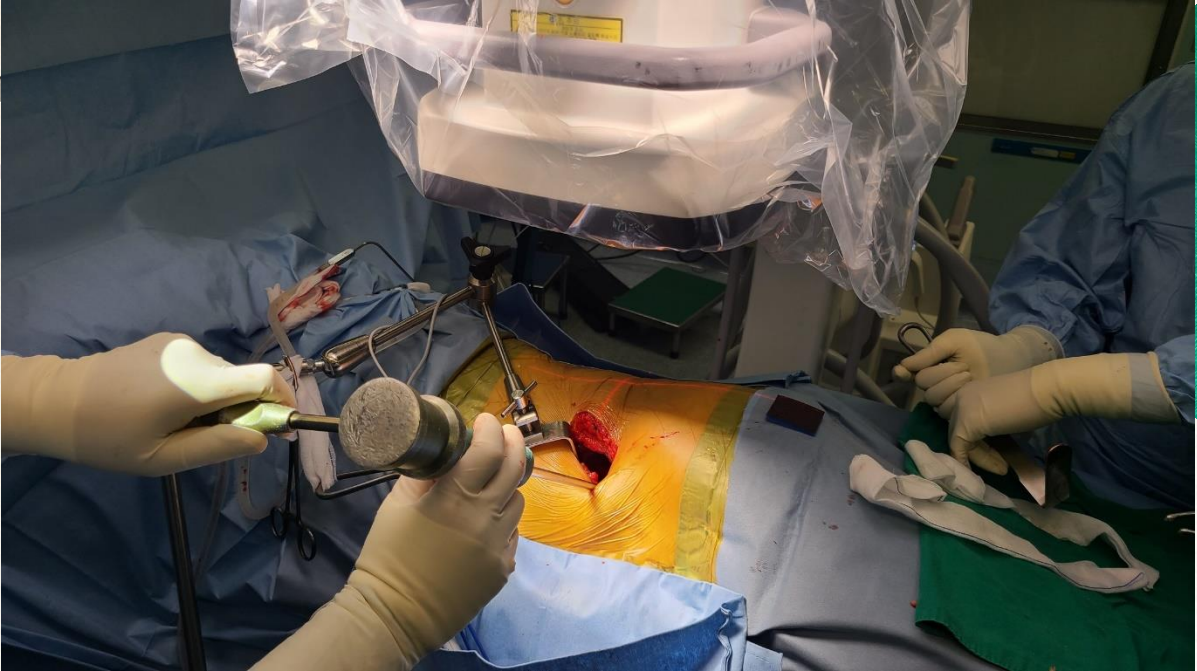
Contracted ALL and anterior annulus
: Tethering effect against making segmental lordosis

ACR (Anterior column realignment)



No special cage, instruments
Oblique insertion of ALIF cage

28 - 31 AUGUST



Considering factors to decide ACR

Target LL

Flexibility of LL

Conventional LLIF vs. ACR

Target LL?

> [Spine \(Phila Pa 1976\)](#). 2022 Dec 15;47(24):1737-1745. doi: 10.1097/BRS.0000000000004449. Epub 2022 Aug 2.

Validation of Age-adjusted Ideal Sagittal Alignment in Terms of Proximal Junctional Failure and Clinical Outcomes in Adult Spinal Deformity

Se-Jun Park¹, Chong-Suh Lee², Byeong-Jik Kang¹, Tae Soo Shin¹, Il Su Kim¹, Jin-Sung Park¹, Keun-Ho Lee³, Dae Hun Shin³

Age-adjusted alignment

: PI-LL = (age-55) /2 + 3

By Lafage et al (Spine, 2017)

> [Neurosurgery](#). 2023 Sep 1;93(3):706-716. doi: 10.1227/neu.0000000000002475. Epub 2023 Mar 31.

A Validation Study of Four Preoperative Surgical Planning Tools for Adult Spinal Deformity Surgery in Proximal Junctional Kyphosis and Clinical Outcomes

Se-Jun Park¹, Chong-Suh Lee², Jin-Sung Park¹, Chung-Youb Jeon¹, Chang-Hyun Ma¹

> [Spine \(Phila Pa 1976\)](#). 2024 Jun 25. doi: 10.1097/BRS.0000000000005077. Online ahead of print.

Ideal Lumbar Lordosis Correction in Patients with Adult Spinal Deformity Without Adversely Impacting Pelvic Tilt and Pelvic Incidence Minus Lumbar Lordosis: A Study of 426 Cases

Se-Jun Park¹, Jin-Sung Park¹, Dong-Ho Kang¹, Chong-Suh Lee²

Treatment of Adult Spinal Deformity with Sagittal Imbalance Using Oblique Lumbar Interbody Fusion: Can We Predict How Much Lordosis Correction Is Possible?

Whoan Jeang Kim, Jae Won Lee, Kun Young Park, Shann Haw Chang, Dae Geon Song, Won Sik Choy

Department of Orthopedic Surgery, Eulji University Hospital, Daejeon, Korea



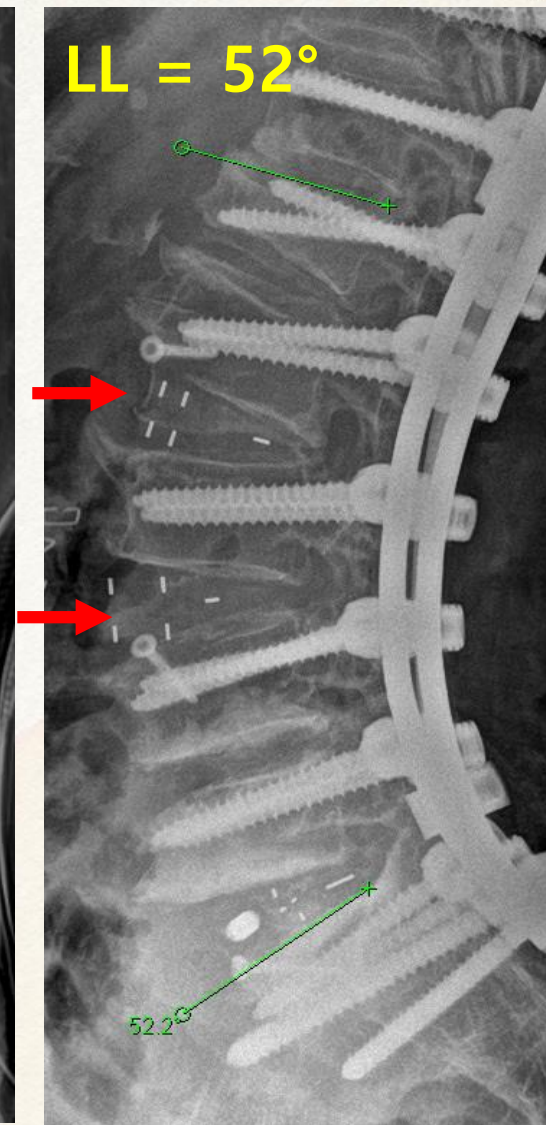
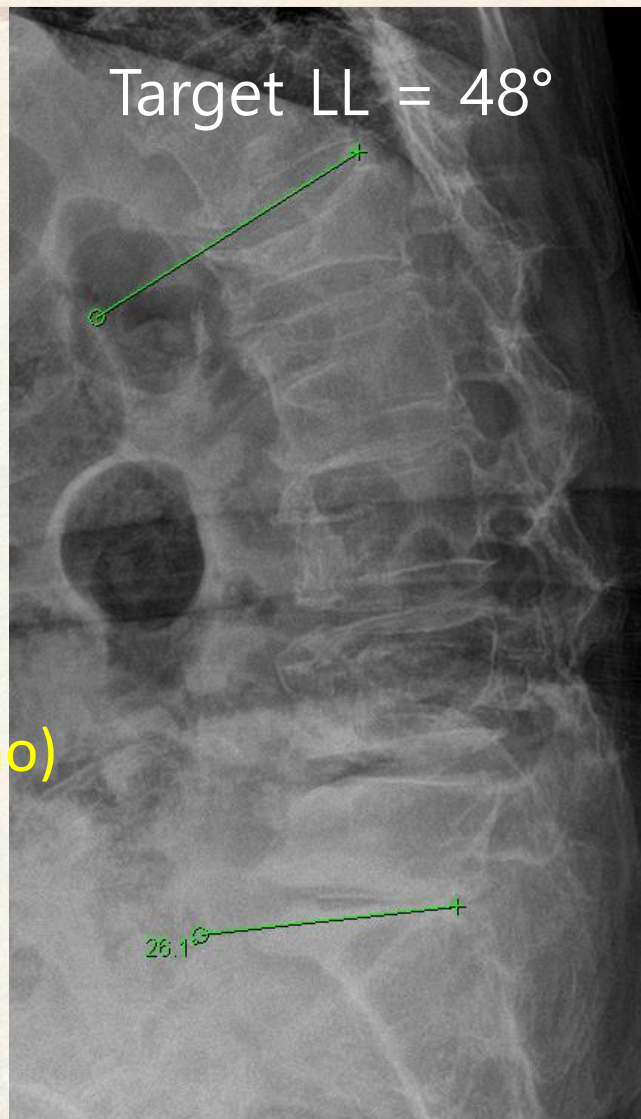
Max. lengthening of anterior disc height in case with ALL-intact procedure

= Max. LL obtainable by conventional LLIF surgery

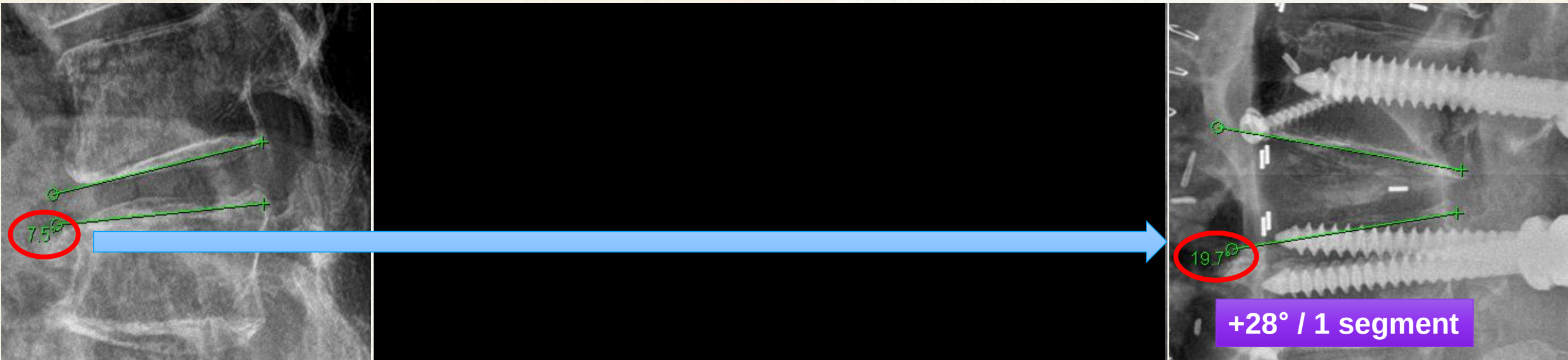


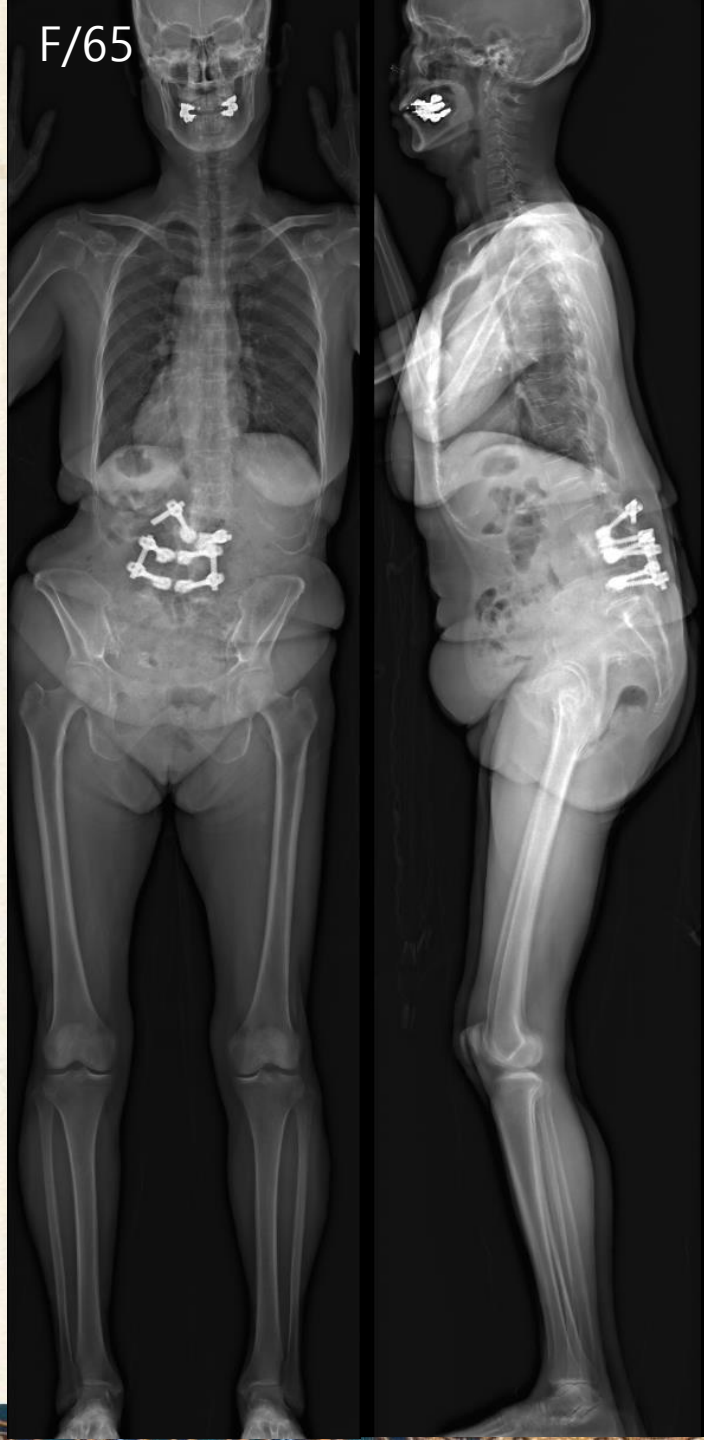
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Serial change of segmental angle during ACR





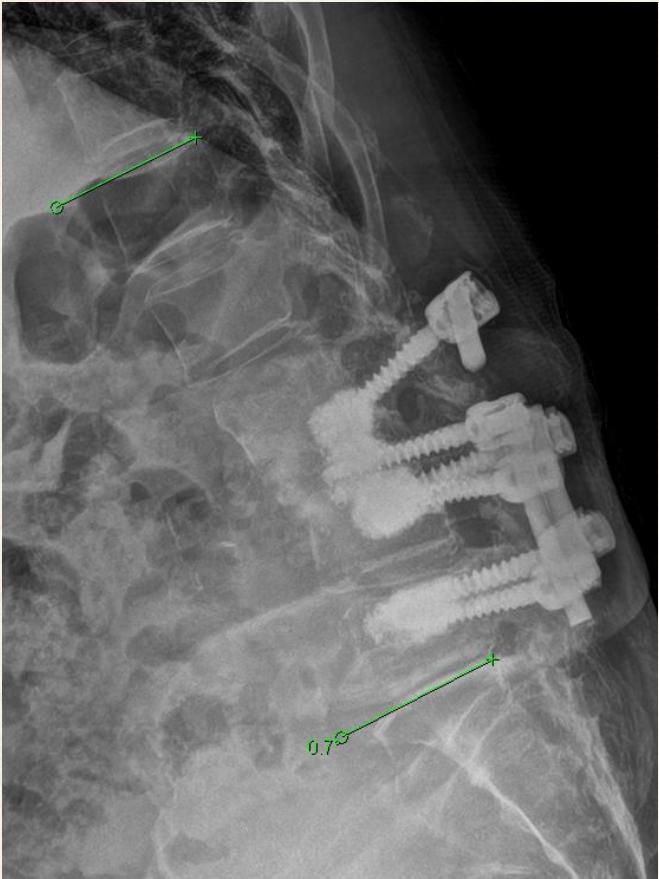
- Intractable back pain with persistent stooping
- s/p fusion surgery @5YA
- $PI = 58^\circ$
- **$LL = 0^\circ$**
- **$PI-LL = 58^\circ$**
- $PT = 35^\circ$
- $SS = 70\text{mm}$

• Target $LL = 51^\circ$ ($46^\circ - 56^\circ$)

→ **ΔLL (target – current LL) = 51°**

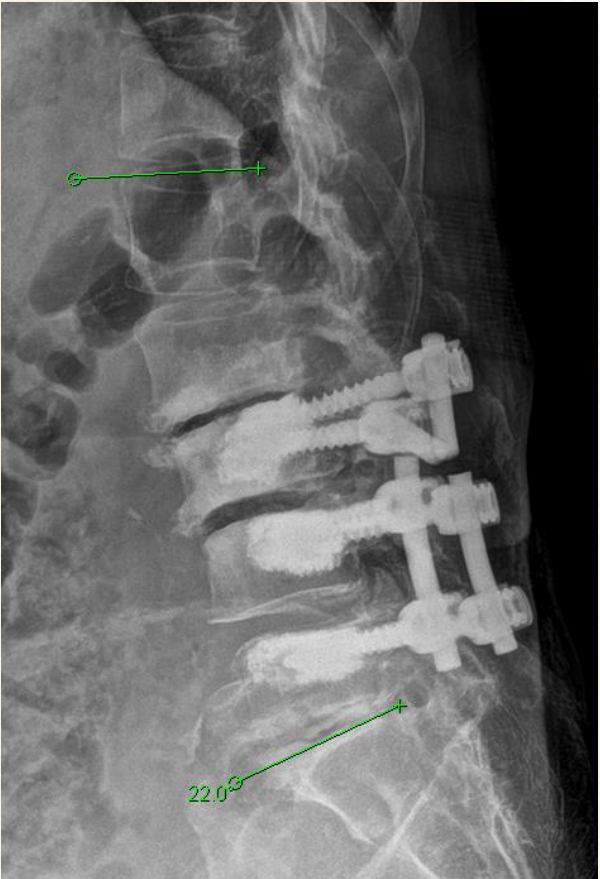
Target LL = 51°

Standing



0°

Extension



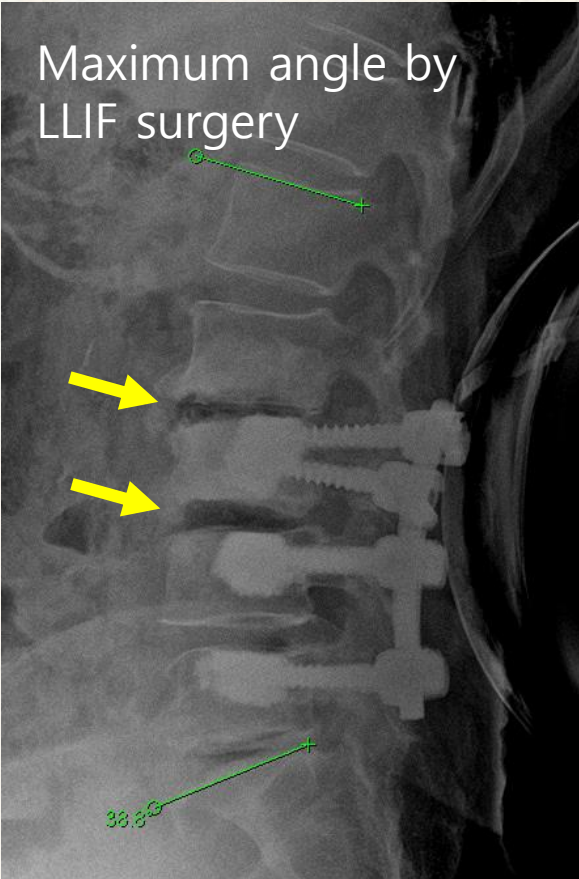
22°

Supine

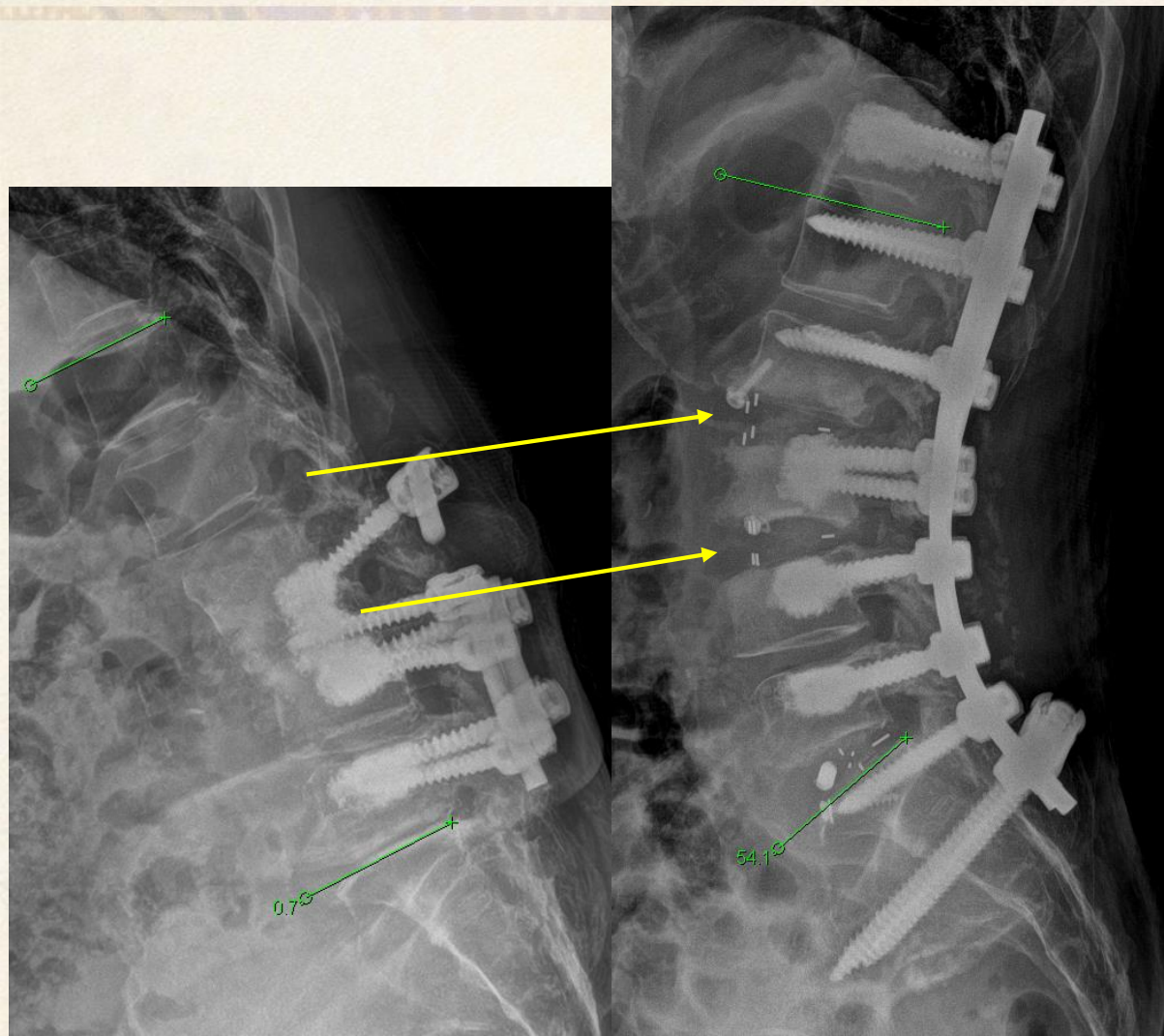


26°

Fulcrum extension



39°



LL=0°

LL=-54°

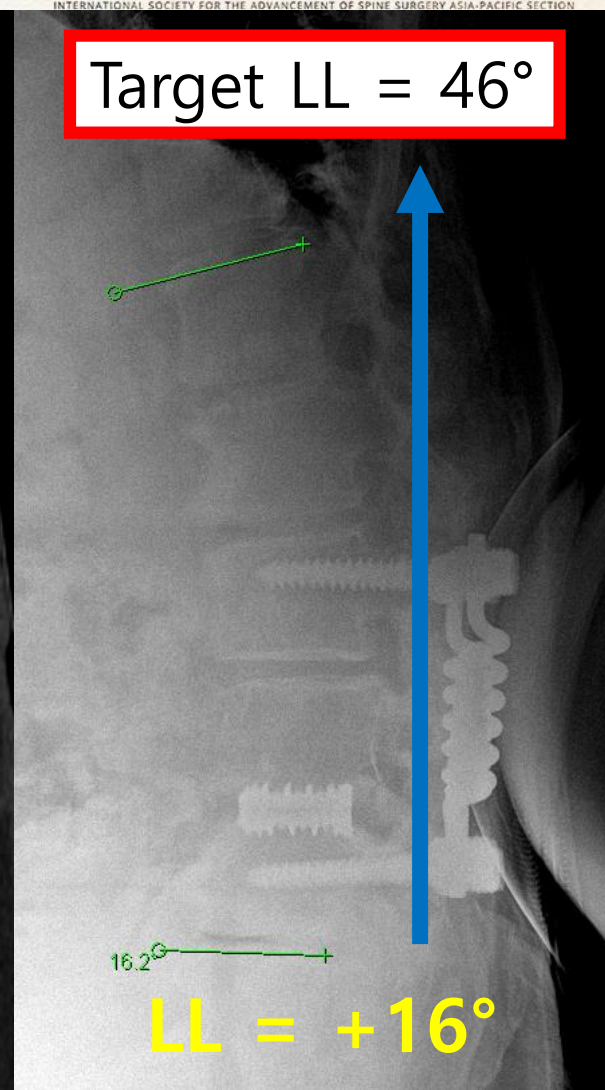
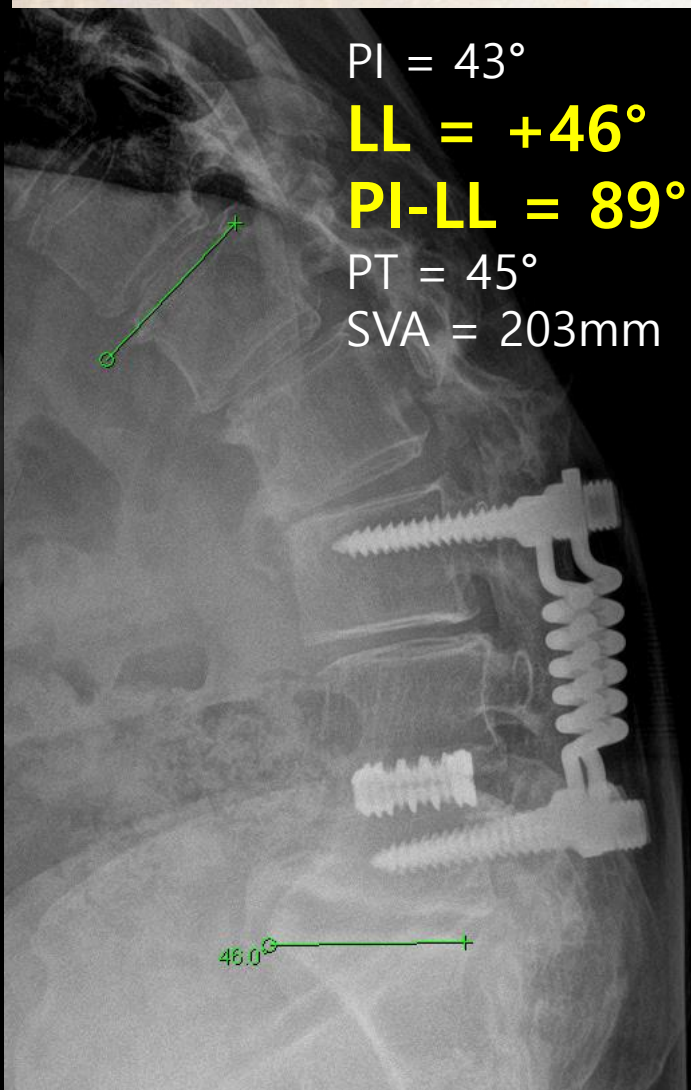


F/64

Walking time < 5min



RTA - INDONESIA

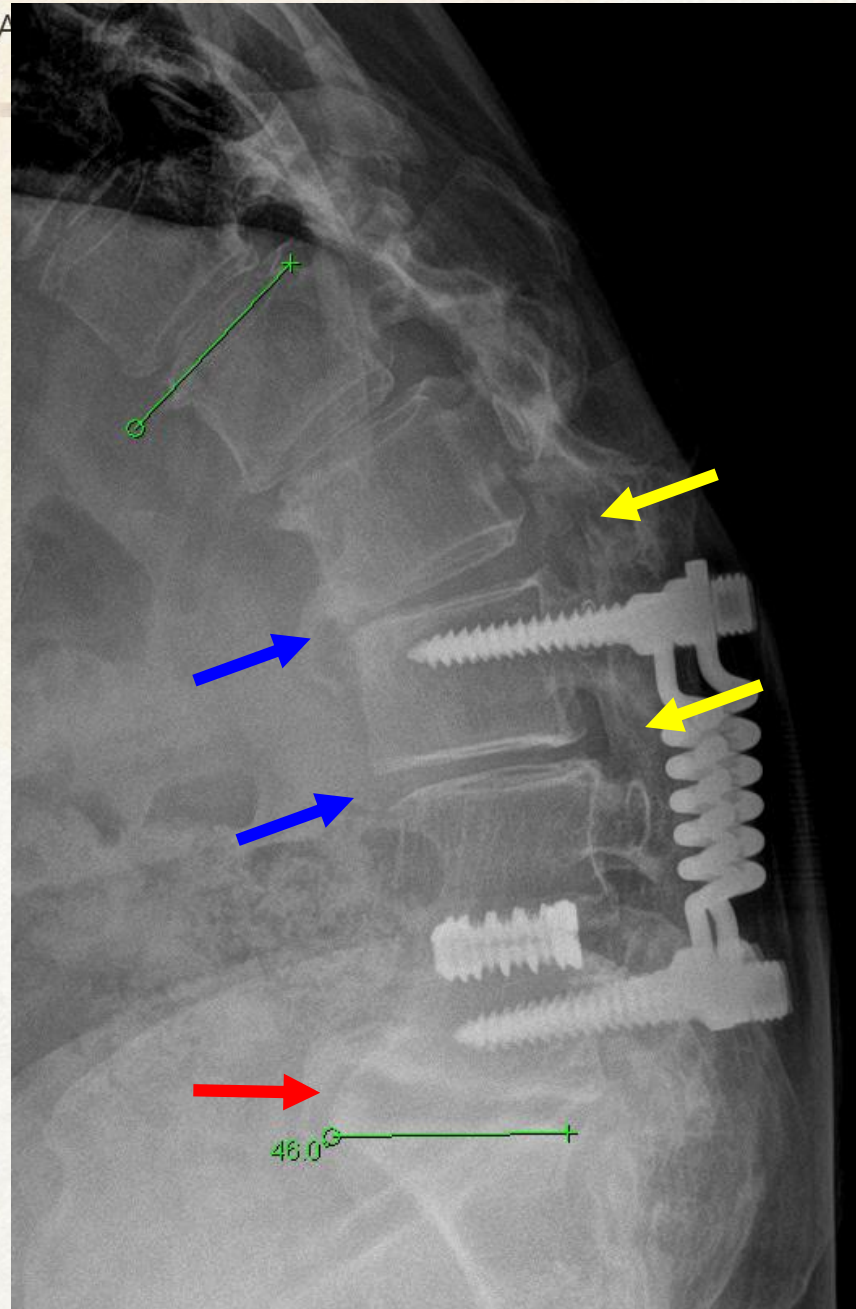


2nd surgery
(2 days apart)

(Lateral surgery)
ACR, L2-3-4

1st surgery

(Anterior surgery)
ALIF, L5-S1



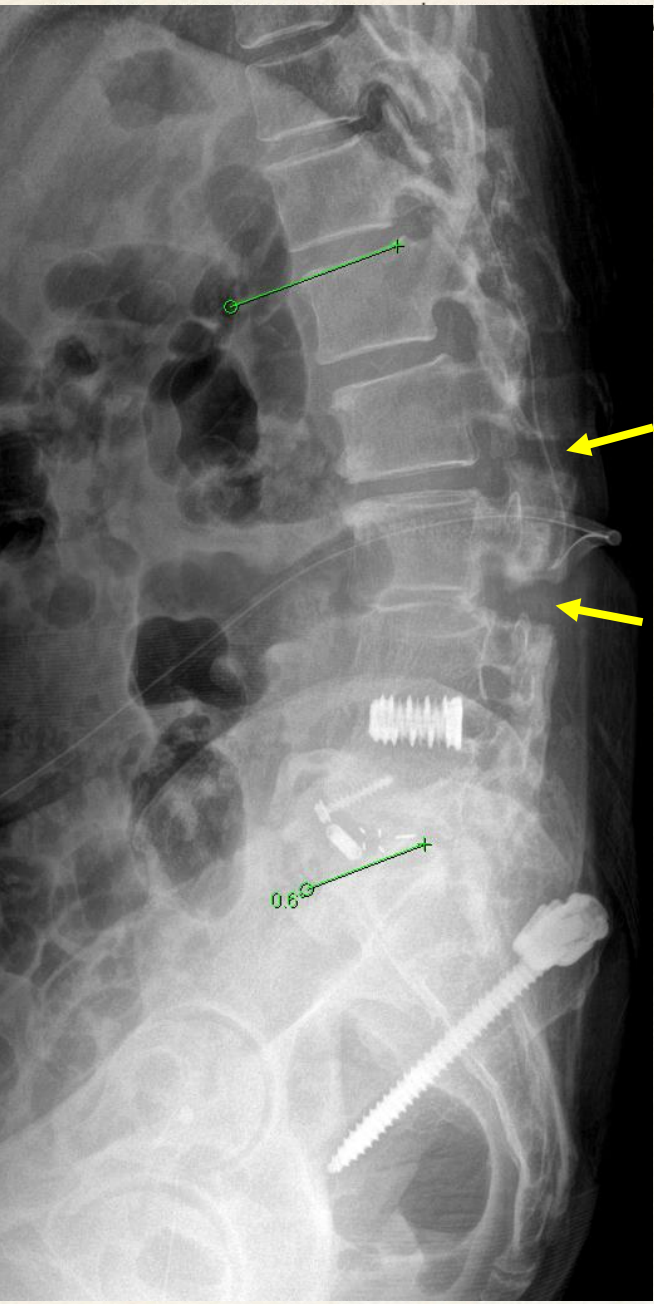
1st surgery

(Posterior surgery)
Implant removal
Ponte osteotomy, L2-3-4
with *fusion mass osteotomy, L3-4*

2nd surgery
(2 days apart)

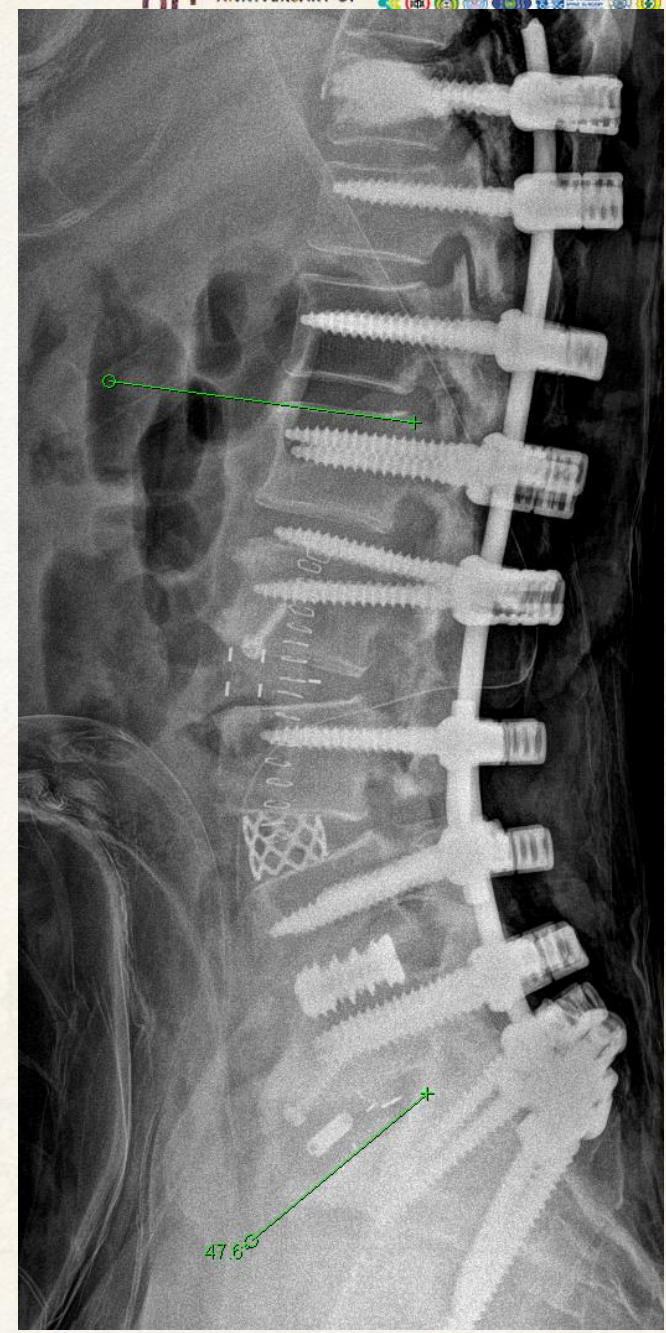
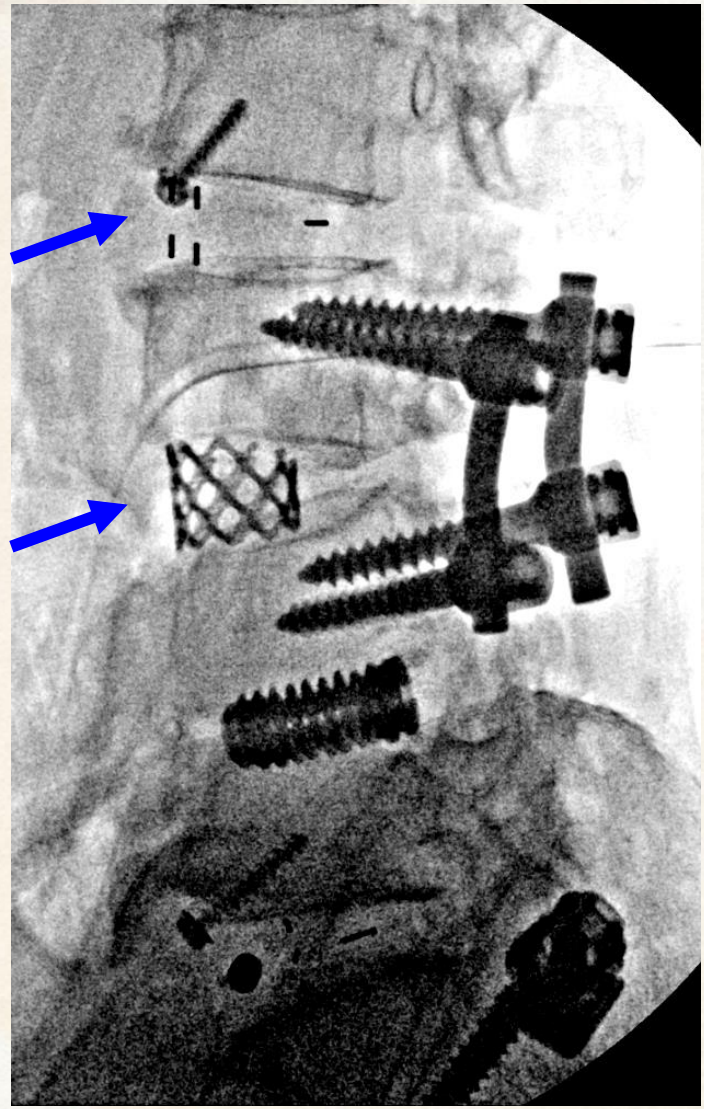
(Posterior surgery)
T10-pelvis Instrumentation

A - INDONESIA

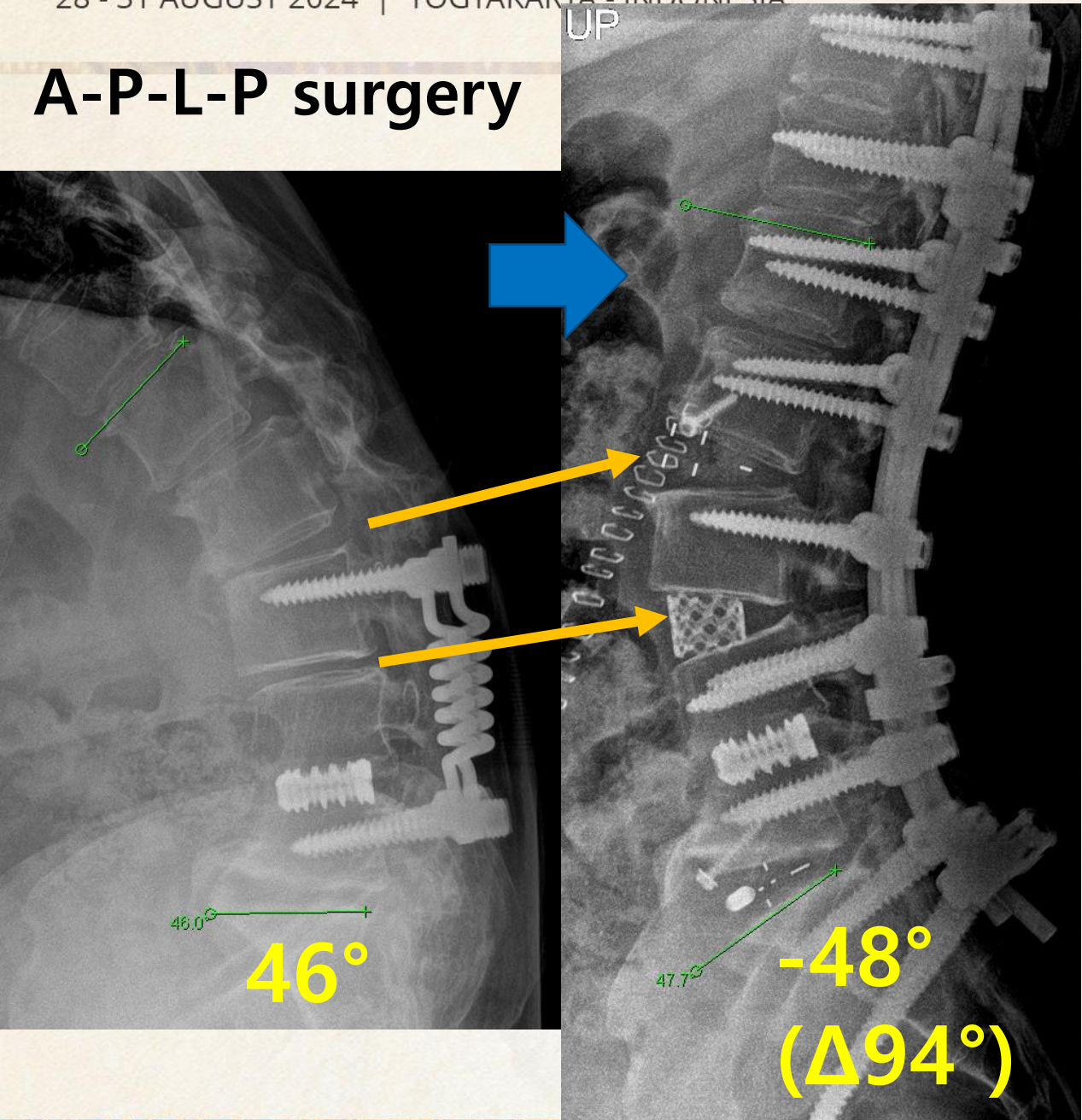


Ponte osteotomy

Ponte osteotomy
(Fusion mass
osteotomy)



A-P-L-P surgery

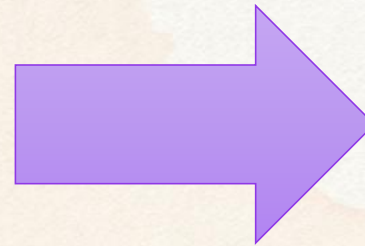




Usefulness of anterior column release for segmental lordosis restoration in degenerative lumbar kyphosis

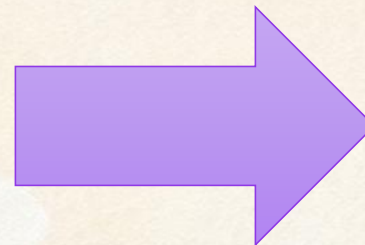
Jin-Sung Park, MD, PhD, Chong-Suh Lee, MD, PhD, Youn-Taek Choi, MD, and Se-Jun Park, MD, PhD

Flexible deformity



Conv. LLIF

Non-flexible deformity



ACR

Thank you for your attention!